

AI, ML and NLP Reference Card Machine Learning * NLP * Azure AI * SHAP * Responsible AI

Data Analyst Career Programme * Module 5 AI-900 * Keep this open while working on your project

ML Classification: Full Pipeline

```
# 1. Encode categoricals
df_model = pd.get_dummies(df[features+[target]],
    columns=['age_band', 'product_type', ...],
    drop_first=True)

# 2. Split with stratify (preserves class balance)
X_train, X_test, y_train, y_test =
    train_test_split(X, y, test_size=0.2,
    stratify=y, random_state=42)

# 3. Scale
scaler = StandardScaler()
X_tr = scaler.fit_transform(X_train)
X_te = scaler.transform(X_test)

# 4. Train (choose one)
lr = LogisticRegression(class_weight='balanced')
rf = RandomForestClassifier(class_weight='balanced')

# 5. Evaluate
preds = model.predict(X_te)
proba = model.predict_proba(X_te)[:,1]
print(classification_report(y_test, preds))
print('AUC:', roc_auc_score(y_test, proba))

# 6. Business cost at threshold
y_t = (proba >= 0.35).astype(int)
tn, fp, fn, tp = confusion_matrix(y_test, y_t).ravel()
cost = fn*800 + fp*25
```

LR vs Random Forest

Model	Pros	Use When
Logistic Regression	Interpretable, fast, coefficients = feature weights	Regulator needs explanations
Random Forest	Higher accuracy, handles non-linear patterns	Performance is primary goal

NLP Path A - Azure Cognitive Services

```
from azure.ai.textanalytics import TextAnalyticsClient
from azure.core.credentials import AzureKeyCredential

client = TextAnalyticsClient(
    endpoint=ENDPOINT,
    credential=AzureKeyCredential(KEY))

# Sentiment (positive/neutral/negative/mixed)
results = client.analyze_sentiment(documents=docs)
for r in results:
    print(r.sentiment,
    r.confidence_scores.positive)

# Key phrases
kp = client.extract_key_phrases(documents=docs)
for r in kp:
    print(r.key_phrases)
```

NLP Path B - Python Open Source

```
# VADER: designed for informal/short text
from vaderSentiment.vaderSentiment import \
    SentimentIntensityAnalyzer
va = SentimentIntensityAnalyzer()

def score(text):
    s = va.polarity_scores(text)
    if s['compound'] >= 0.05: return 'positive'
    elif s['compound'] <= -0.05: return 'negative'
    else: return 'neutral'

# spaCy: entity + key phrase extraction
import spacy
nlp = spacy.load('en_core_web_sm')

def extract(text):
    doc = nlp(text)
    ents = [(e.text, e.label_) for e in doc.ents]
    phrases = [c.text for c in doc.noun_chunks]
    return ents, phrases
```

SHAP Explainability

```
import shap

# Tree model (Random Forest)
explainer = shap.TreeExplainer(rf_model)
shap_values = explainer.shap_values(X_te)
sv = shap_values[1] # class 1 = escalated

# Global: what drives escalation?
shap.summary_plot(sv, X_te,
    feature_names=X.columns.tolist(),
    plot_type='bar')

# Individual: why was this customer flagged?
shap.waterfall_plot(
    shap.Explanation(
        values=sv[idx],
        base_values=explainer.expected_value[1],
        data=X_te[idx],
        feature_names=X.columns.tolist()
    ))

# Linear model (Logistic Regression)
explainer = shap.LinearExplainer(lr_model, X_tr)
sv = explainer.shap_values(X_te)
```

Responsible AI Checklist

- * State intended use AND prohibited uses
- * Identify protected characteristic proxies in features
- * Check FPR by demographic group - not just overall accuracy
- * Run recall check per group: which group does the model miss most?
- * SHAP explanation available for every individual prediction
- * Human-in-the-loop process documented before deployment
- * Quarterly bias monitoring process defined

Evaluation for Imbalanced Datasets

Metric	Formula	Use When
--------	---------	----------

Azure AI-900: Service Quick Reference

Azure Service	What It Does	API Capability
---------------	--------------	----------------

Power BI AI-Native Features

Feature	Where	Best For
---------	-------	----------

Recall (TPR)	TP / (TP+FN)	Missing high-risk cases is costly (FCA complaint)
Precision	TP / (TP+FP)	Unnecessary interventions are costly
F1 Score	2*P*R / (P+R)	Both errors matter equally
AUC	Area under ROC curve	Overall discrimination, threshold-independent
Business Cost	FN*cost_FN + FP*cost_FP	When the two error costs are known

Accuracy is misleading on imbalanced data.

If 73% of customers did not escalate, a model that always predicts 'not escalated' would show 73% accuracy. Always check recall and AUC on imbalanced classes.

Bias Audit Pattern

```

for group in df_test['age_band'].unique():
    s = df_test[df_test['age_band']==group]
    neg = s[s['y_true']==0]
    fpr = (neg['y_pred']==1).mean()
    rec = (s[s['y_true']==1]['y_pred']==1).mean()
    print(f'{group}: FPR={fpr:.3f} Recall={rec:.3f}')

```

Azure AI Language	NLP: sentiment, key phrases, entities, language detection	analyze_sentiment, extract_key_phrases, recognize_entities
Azure OpenAI	Generative AI: text completion, summarisation, Q and A	chat.completions (GPT-4)
Azure Computer Vision	Image analysis: objects, text (OCR), faces, scenes	analyze_image, read (OCR)
Azure Speech	Speech to text and text to speech	recognize_once, synthesize_speech
Azure ML Studio	Build, train, deploy ML models in the cloud	Designer (no-code) + Python SDK

Free tier (F0) limits:

- * Azure AI Language: 5,000 text records per month
- * Azure OpenAI: limited credits on free trial
- * Azure ML: free compute cluster hours
- * Always create resources in UK South or West Europe for GDPR

Key Influencers	Visualisations pane	What drives a metric? Automatic ML insight.
Decomposition Tree	Visualisations pane	Drill down through multiple dimensions interactively
Q and A	Insert > Q and A	Natural language queries for non-technical users
Smart Narrative	Visualisations pane	Auto-generated text summary of a visual
Anomaly Detection	Line chart format pane	Spot unusual data points automatically

Submission Checklist

Section A - Framing

- * Intended use, prohibited uses, protected characteristics, human-in-the-loop, error costs

Section C - ML Model

- * Two models compared, business cost at three thresholds, threshold justified

Section D - NLP

- * All five NLP outputs built, Path A or B documented

Section E - Bias Audit

- * FPR and recall by age band and product type, deployment judgement written

Section F - SHAP

- * Global SHAP chart, 3 waterfall charts (TP, FP, FN), 3 regulator explanations

Section H - Documentation

- * AI Reflection 300 words, Responsible AI Summary one page